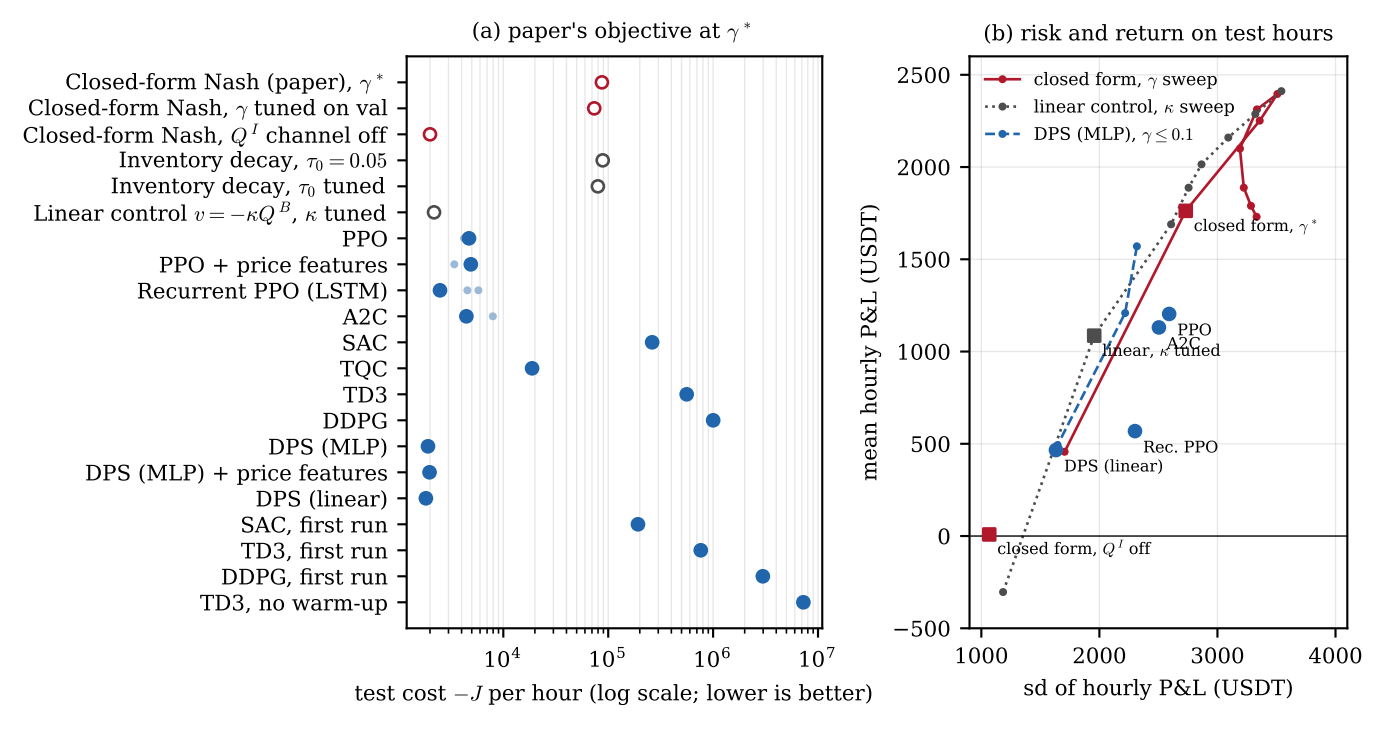
(a) paper's objective at γ^*
Closed-form Nash (paper), γ^*
Closed-form Nash, γ tuned on val
Closed-form Nash, Q^I channel off
Inventory decay, $\tau_0 = 0.05$
Inventory decay, τ_0 tuned
Linear control $v = -\kappa Q^B$, κ tuned
PPO
PPO + price features
Recurrent PPO (LSTM)
A2C
SAC
TQC
TD3
DDPG
DPS (MLP)
DPS (MLP) + price features
DPS (linear)
SAC, first run
TD3, first run
DDPG, first run
TD3, no warm-up
10^4
10^5
10^6
10^7
test cost $-J$ per hour (log scale; lower is better)
(b) risk and return on test hours
closed form, γ sweep
linear control, κ sweep
DPS (MLP), $\gamma \le 0.1$
closed form, γ^*
PPO
A2C
linear, κ tuned
Rec. PPO
DPS (linear)
closed form, Q^I off
mean hourly P&L (USDT)
sd of hourly P&L (USDT)
2500
2000
1500
1000
500
0
−500
1000
2000
3000
4000